MILITARY SPECIFICATION SHEET

BOLT, SLEEVE, 100° TENSION FLUSH HEAD, STRAIGHT SHANK, NICKEL-CRES 304 SLEEVE, NICKEL-ALLOY 718 PIN, 220 KSI Ftu, 450°F

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-85667.

1. Application criteria. The sleeve bolt shall be used to provide a precision net fit or controlled interference fit. The sleeve bolts shall be used in accordance with the limitations of MIL-B-85667/1(AS). The sleeve bolts shall not be degreased and cetyl alcohol shall not be removed before installation.

2. Companion and accessory parts. Nuts in accordance with MIL-N-85730 shall be used with the sleeve bolts.

3. Oversize bolts. First and second oversize sleeve bolts may replace MIL-B-85667/11 sleeve bolts for first and second repair operations.

4. Installation. Installation shall be in accordance with MIL-HDBK-271-104.

5. Materials.

5.1 Pin. Pin material shall be nickel-alloy 718 in accordance with AMS 5662 (UNS N07718).

5.2 Cold work. Pin shall be cold worked and aged to 220 KSI Ftu and 125 KSI Fsu.

5.3 Sleeve. Sleeve material shall be corrosion resistant steel, AISI 304 (UNS S30400), in accordance with MIL-T-8504.

6. Heat treatment.

6.1 Pin. The pin shall be heat treated in accordance with AMS 5662.

AMSC N/A FSC 5306
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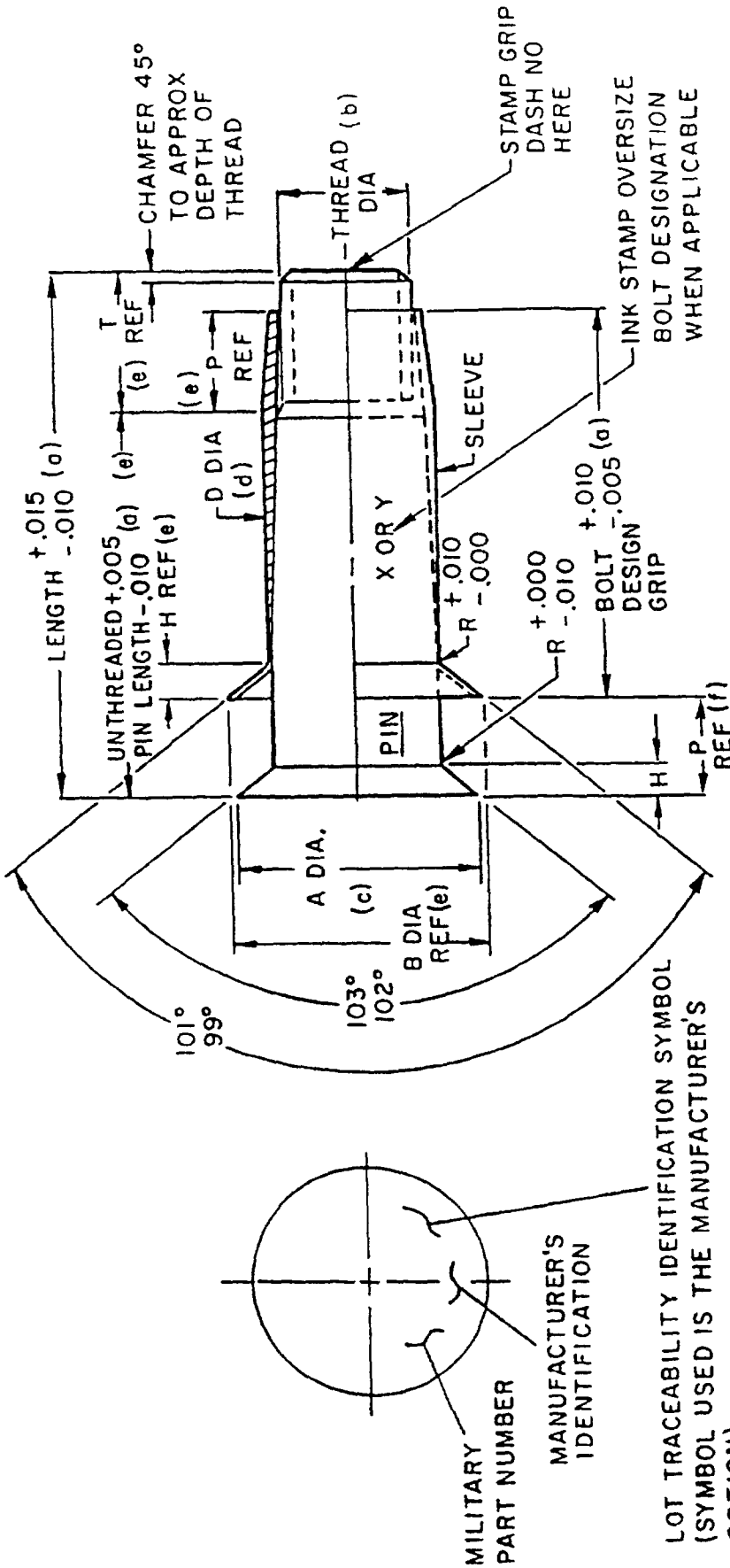


FIGURE 1. Tension flush head bolt.

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NOTES.

a. Grip lengths are available in .0625 increments.

b. Threads shall be in accordance with MIL-S-8879. Thread major diameter (T.D.) shall be in accordance with Table I.

c. 1. For sizes .190 thru .375, dimension "A" shall be concentric to dimension "D" within .008 FIM.

2. For sizes .437 thru .625, .012 FIM.

3. For size .750, .020 FIM.

d. 1. For sizes .190 thru .375, dimension "D" shall be concentric to thread pitch diameter within .006 FIM.

2. For sizes .437 thru 0.750, .010 FIM.

e. Reference dimensions are for design purposes only and are not an inspection or manufacturing requirement.

f. Protrusion will vary according to taper to provide expansion as given in Table I.

6.1.1 Hardness. Hardness shall not be greater than 50 HRC.

6.2 Sleeve. Sleeve is cold worked.

7. Surface texture. Surface texture shall be R_a-32 in accordance with ANSI/ASME B46.1.

8. Coatings or surface treatment.

8.1 Pin. The pin shall be dry-film lubricated in accordance with MIL-L-46010, Type I, .0003/.0005 thick.

8.2 Sleeve. None.

9. Lubricants. 0.0001-inch overspray allowed on head and threads. After application of dry lubrication, the pin outside diameter and sleeve inside diameter shall be lubricated with cetyl alconol in accordance with MIL-L-6715.

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TABLE I. Bolt dimensions.

Size number	Grip	Dash numbers	Thread Per MIL-S-8879 UNJF-3A	A bolt head dia	B Ref sleeve flange Diameter Std	B Ref sleeve flange Diameter X O/S	B Ref sleeve flange Diameter Y O/S	H	Fm Flange height Ref	R Ref	Fillet Radius	T.O Thread dia	T Thread length	J Dimension Initial Sleeve Diameter (a) Std	J Dimension Initial Sleeve Diameter (a) X O/S	J Dimension Initial Sleeve Diameter (a) Y O/S	D Dimension Final Sleeve Diameter (a)(b) Std	D Dimension Final Sleeve Diameter (a)(b) X O/S	D Dimension Final Sleeve Diameter (a)(b) Y O/S	Minimum Installation Hole Size Diameter Std	Minimum Installation Hole Size Diameter X O/S	Minimum Installation Hole Size Diameter Y O/S
3	1	7 thru 10	1900-32	.416 .369	.457	.463	.469	.083 .081	.093 .095	.225 .432	.030	.1840 .1810	.343	1265 2360	2425 2420	2485 2480	2455 2445	2515 2505	2575 2565	.237	.243	.249
	2	11 thru 16		.420 .370	.459	.465	.471		.093	.432				2365 2360	2425 2420	2485 2480	2455 2445	2515 2505	2575 2565	.237	.243	.249
	3	17 thru -22		.430 .380	.476	.483	.490		.095	.432				2485 2480	2555 2550	2625 2620	2575 2565	2645 2635	2715 2705	.249	.256	.263
4	1	8 thru -10	2500-28	.550 .494	.586	.597	.603	.110 .108	.120 .120	.225 .432	.030	.2440 .2410	.406	3005 3000	3115 3110	3175 3170	3095 3085	3205 3195	3265 3255	.301	.312	.318
	2	-11 thru -17		.553 .496	.588	.599	.605							3005 3000	3115 3110	3175 3170	3095 3085	3205 3195	3265 3255	.301	.312	.318
	3	18 thru -28		.566 .515	.607	.614	.623		.123	.432				3145 3140	3215 3210	3305 3300	3235 3225	3305 3295	3395 3385	.315	.322	.331
5	1	-8 thru -11	3125-24	.684 .618	.717	.724	.731	.138 .136	.150 .148	.225 .432	.040	.3060 .3020	.484	3665 3660	3735 3730	3805 3800	3755 3745	3825 3815	3895 3885	.367	.374	.381
	2	12 thru -20		.688 .622	.719	.726	.733							3665 3660	3735 3730	3805 3800	3755 3745	3825 3815	3895 3885	.367	.374	.381
	3	-21 thru -31		.702 .636	.740	.748	.755		.151	.432				3805 3800	3885 3880	3955 3950	3895 3885	3975 .3965	4045 4035	.381	.389	.396
6	1	-9 thru -11	3750-24	.819 .746	.848	.856	.868	.166 .164	.175 .176	.225 .432	.040	.3660 .3640	.562	4315 4310	4395 4390	4515 4510	4405 4395	4485 4475	4605 4595	.432	.440	.452
	2	-12 thru -22		.823 .750	.851	.859	.871							4315 4310	4395 4390	4515 4510	4405 4395	4485 4475	4605 4595	.432	.440	.452
	3	-23 thru -34		.838 .765	.872	.880	.888	---	.178	.432				4475 4470	4555 4550	4635 4630	4565 4555	4645 4635	4725 4715	.448	.456	.464
7	1	-9 thru -12	4375-20	.956 .875	.987	.999	1.007	.194 .192	.205 .207	.225 .432	.050	.4310 .4260	.625	.4985 .4980	.5105 .5100	.5185 .5180	.5075 .5065	.5195 .5185	.5275 .5265	.499	.511	.519
	2	-13 thru -26		.960 .878	.991	1.003	1.011							4985 4980	5105 5100	.5185 5180	5075 5065	.5195 5185	.5275 5265	499	511	519
	3	-27 thru -40		.978 .895	1.015	1.026	1.038		.208	.432				5185 5180	5295 5290	5415 5410	.5275 5265	5385 5375	5505 5495	519	530	542
8	1	-10 thru -12	5000-20	1.093 1.005	1.120	1.130	1.140	.222 .220	.231 .233	.225 .432	.050	.4930 .4880	.703	.5695 5690	5795 5790	5895 5890	5785 5775	5885 5875	5985 .5975	570	580	590
	2	-13 thru -30		1.097 1.008	1.125	1.135	1.145							5695 5690	5795 5790	5895 5890	5785 5775	.5885 5875	.5985 5975	570	580	590
	3	-31 thru -48		1.118 1.035	1.152	1.162	1.171		.236	.432				5895 5890	5995 5990	6085 6080	5985 5975	6085 6075	.6175 6165	590	600	609
9	1	-14 thru -23	5625-18	1.195 1.100	1.232	1.247	1.260	.240 .238	.249 .253	.432 .432	.070	.5550 .5500	.781	6395 6390	6545 6540	6675 6670	6485 6475	6635 6625	.6765 .6755	640	655	669
	2	-24 thru -48		1.226 1.131	1.274	1.290	1.305							6705 6700	6865 6860	.7015 7010	.6795 6785	6955 6945	.7105 .7095	671	.687	702
	3	-49 thru -72		1.258 1.161	1.316	1.332	1.347		.258	.900				7015 7010	7175 7170	.7325 7320	.7105 .7095	7265 7255	.7415 7405	702	718	733
10	1	-14 thru -25	6250-18	1.298 1.195	1.355	1.374	1.394	.255 .253	.272 .270	.432 .432	.073	.6180 .6120	.843	7075 7070	7265 7260	7465 7460	7165 7155	7355 .7345	7555 7545	708	727	747
	2	-26 thru -52		1.332 1.231	1.389	1.407	1.422							7465 7460	7645 .7640	7795 7790	7555 7545	.7735 7725	.7885 7875	747	765	780
	3	-53 thru -78		1.366 1.264	1.432	1.448	1.464		.274	.900				7795 7790	.7955 7950	.8115 8110	7885 7875	8045 .8035	.8205 8195	.780	.796	812
12	1	-14 thru -29	7500-16	1.492 1.389	1.549	1.564	1.580	.280 .278	.296 .295	.432 .432	.073	.7430 .7370	1.000	8425 8420	8575 8570	8735 8730	8515 8505	8665 8655	.8825 8815	843	858	874
	2	-30 thru -60		1.531 1.417	1.588	1.608	1.624							8845 8840	9045 9040	9205 9200	8935 9125	9135 9125	9295 9285	885	905	921
	3	-61 thru -90		1.570 1.455	1.655	1.671	1.686		.308	.900				9205 9200	9365 9360	9515 9510	9295 9285	9455 9445	9605 9595	921	937	952

NOTES:

(A) D dia. shall be verified within the grip bearing portion engaging the bolt shank prior to expansion; when bolt is installed to sleeve flange the dia. shall be within the expanded D dia. Limits shown in Table I.

(B) The sleevebolt is manufactured as a precision assembly and the components should never be dismantled. In the event of inadvertent separation, please discard and replace with a new assembly.

(C) All dimensions in inches, tolerances: decimals ±.010", angles ±2° unless otherwise specified.

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TABLE II. Mechanical properties.

Dia Dash Number	Ultimate Tensile Strength Lbs-Min (b)	Thread Area In^2 (b)	Double Shear Strength Lbs-Min (a)	Pin Test Diameter (+.0005) In. (a)	Pin Test Dia Area In^2 (a)	Max Install Torque In.-Lbs	Fatigue Test Loading Lbs ± 2% (c)	
							Max	Min
-3	4,970	.0217	7,080	.1900	.0284	60	2,140	214
-4	8,870	.0388	12,300	.2500	.0491	115	3,800	380
-5	14,100	.0614	19,200	.3125	.0767	185	6,000	600
-6	20,900	.0950	27,600	.3750	.1104	335	9,350	935
-7	28,300	.1288	37,500	.4375	.1503	540	12,700	1,270
-8	37,800	.1717	49,200	.5000	.1963	805	16,900	1,690
-9	47,900	.2176	62,200	.5625	.2485	1080	21,500	2,150
-10	59,700	.2724	76,000	.6250	.3068	1560	26,900	2,690
-12	86,900	.3952	110,000	.7500	.4418	2930	39,000	3,900

Notes:

a. Shear test performed on pins ground to pin test diameter. The shear values are based on a minimum shear strength of 125 KSI and the stress area at the test diameter.

b. The ultimate tensile strength values are calculated on the basic pitch diameter of the external thread and a tensile strength of 220 KSI.

c. Maximum fatigue load is based on 46% of the ultimate tensile strength. Minimum fatigue load is 10% of maximum fatigue load.

10. Example of military part number.

M85667/11 - 12 - 34

- 34 — Grip length dash number in sixteenths of an inch. (Stamp on threaded end.)
- '-' designates basic shank diameter. X designates 1st oversize, Y designates 2nd oversize. (Ink stamp on sleeve when applicable.)
- 12 — Diameter dash numbers in sixteenths of an inch.
- M85667/11 — Basic specification sheet number.

11. Administrative and contractual provisions.

a. For design purposes and in case of conflict, this document takes precedence over documents referenced herein.

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TABLE III. Bolt grips and lengths.

Grip Number	Bolt Design Grip	-3 Diameter		-4 Diameter		-5 Diameter	
		Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length
6	.375	.405	.748	.405	.811	.405	.889
7	.438	.468	.811	.468	.874	.468	.952
8	.500	.530	.873	.530	.936	.530	1.014
9	.562	.592	.935	.592	.998	.592	1.076
10	.625	.655	.998	.655	1.061	.655	1.139
11	.688	.718	1.061	.718	1.124	.718	1.202
12	.750	.780	1.123	.780	1.186	.780	1.264
13	.812	.842	1.185	.842	1.248	.842	1.326
14	.875	.905	1.248	.905	1.311	.905	1.389
15	.938	.968	1.311	.968	1.374	.968	1.452
16	1.000	1.030	1.373	1.030	1.436	1.030	1.514
17	1.062	1.092	1.435	1.092	1.498	1.092	1.576
18	1.125	1.155	1.498	1.155	1.561	1.155	1.639
19	1.188	1.218	1.561	1.218	1.624	1.218	1.702
20	1.250	1.280	1.623	1.280	1.686	1.280	1.764
21	1.312	1.234	1.685	1.234	1.748	1.342	1.826
22	1.375	1.405	1.748	1.405	1.811	1.405	1.889
23	1.438	1.468	1.811	1.468	1.874	1.468	1.952
24	1.500	1.530	1.873	1.530	1.936	1.530	2.014
25	1.562			1.592	1.998	1.592	2.076
26	1.625			1.655	2.061	1.655	2.139
27	1.688			1.718	2.124	1.718	2.202
28	1.750			1.780	2.186	1.780	2.264
29	1.812					1.842	2.326
30	1.875					1.905	2.389
31	1.938					1.968	2.452

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TABLE III. Bolt grips and lengths (continued).

Grip Number	Bolt Design Grip	-6 Diameter		-7 Diameter		-8 Diameter	
		Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length
6	.375	.405	.968	.405	1.030	.405	1.108
7	.438	.468	1.030	.468	1.093	.468	1.171
8	.500	.530	1.092	.530	1.155	.530	1.233
9	.562	.592	1.154	.592	1.217	.592	1.295
10	.625	.655	1.217	.655	1.280	.655	1.358
11	.688	.718	1.280	.718	1.343	.718	1.421
12	.750	.780	1.342	.780	1.405	.780	1.483
13	.812	.842	1.404	.842	1.467	.842	1.545
14	.875	.905	1.467	.905	1.530	.905	1.608
15	.938	.968	1.530	.968	1.593	.968	1.671
16	1.000	1.030	1.592	1.030	1.655	1.030	1.733
17	1.062	1.092	1.654	1.092	1.717	1.092	1.795
18	1.125	1.155	1.717	1.155	1.780	1.155	1.858
19	1.188	1.218	1.780	1.218	1.843	1.218	1.921
20	1.250	1.280	1.842	1.280	1.905	1.280	1.983
21	1.312	1.342	1.904	1.342	1.967	1.342	2.045
22	1.375	1.405	1.967	1.405	2.030	1.405	2.108
23	1.438	1.468	2.030	1.468	2.093	1.468	2.171
24	1.500	1.530	2.092	1.530	2.155	1.530	2.233
25	1.562	1.592	2.154	1.592	2.217	1.592	2.295
26	1.625	1.655	2.217	1.655	2.280	1.655	2.358
27	1.688	1.718	2.280	1.718	2.343	1.718	2.421
28	1.750	1.780	2.342	1.780	2.405	1.780	2.483
29	1.812	1.842	2.404	1.842	2.467	1.842	2.545
30	1.875	1.905	2.467	1.905	2.530	1.905	2.608
31	1.938	1.968	2.530	1.968	2.593	1.968	2.671
32	2.000	2.030	2.592	2.030	2.655	2.030	2.733
33	2.062	2.092	2.654	2.092	2.717	2.092	2.795
34	2.125	2.155	2.717	2.155	2.780	2.155	2.858
35	2.188			2.218	2.843	2.218	2.921
36	2.250			2.280	2.905	2.280	2.983
37	2.312			2.342	2.967	2.342	3.045
38	2.375			2.405	3.030	2.405	3.108
39	2.438			2.468	3.093	2.468	3.171
40	2.500			2.530	3.155	2.530	3.233
41	2.562					2.592	3.295
42	2.625					2.655	3.358
43	2.688					2.718	3.421
44	2.750					2.780	3.483
45	2.812					2.842	3.545
46	2.875					2.905	3.608

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TABLE III. Bolt grips and lengths (continued).

Grip Number	-9 Diameter		-10 Diameter		-12 Diameter	
	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length
14	.905	1.686	.905	1.748	.905	1.905
15	.968	1.748	.968	1.811	.968	1.966
16	1.030	1.811	1.030	1.873	1.030	2.030
17	1.092	1.874	1.092	1.935	1.092	2.092
18	1.155	1.936	1.155	1.998	1.155	2.155
19	1.218	1.998	1.218	2.061	1.218	2.218
20	1.280	2.061	1.280	2.123	1.280	2.280
21	1.342	2.124	1.342	2.185	1.342	2.342
22	1.405	2.188	1.405	2.248	1.405	2.405
23	1.468	2.248	1.468	2.311	1.468	2.468
24	1.530	2.311	1.530	2.373	1.530	2.530
25	1.592	2.374	1.592	2.435	1.592	2.592
26	1.655	2.436	1.655	2.498	1.655	2.655
27	1.718	2.498	1.718	2.561	1.718	2.718
28	1.780	2.561	1.780	2.623	1.780	2.780
29	1.842	2.624	1.842	2.685	1.842	2.842
30	1.905	2.686	1.905	2.748	1.905	2.905
31	1.968	2.748	1.968	2.811	1.968	2.968
32	2.030	2.811	2.030	2.873	2.030	3.030
33	2.092	2.874	2.092	2.935	2.092	3.092
34	2.155	2.936	2.155	2.998	2.156	3.155
35	2.218	2.998	2.218	3.061	2.218	3.218
36	2.280	3.061	2.280	3.123	2.280	3.280
37	2.342	3.124	2.342	3.185	2.342	3.342
38	2.405	3.186	2.405	3.248	2.405	3.405
39	2.468	3.248	2.468	3.311	2.468	3.468
40	2.530	3.311	2.530	3.373	2.530	3.530
41	2.592	3.374	2.592	3.435	2.592	3.592
42	2.655	3.436	2.655	3.498	2.655	3.655
43	2.718	3.498	2.718	3.561	2.718	3.718
44	2.780	3.561	2.780	3.623	2.780	3.780
45	2.842	3.624	2.842	3.685	2.842	3.842
46	2.905	3.686	2.905	3.748	2.905	3.905
47	2.968	3.748	2.968	3.811	2.968	3.968
48	3.030	3.811	3.030	3.873	3.030	4.030
49	3.092	3.874	3.092	3.935	3.092	4.092
50	3.155	3.896	3.155	3.998	3.155	4.155
51	3.218	3.998	3.218	4.061	3.218	4.218
52	3.280	4.061	3.280	4.123	3.280	4.280

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TABLE III. Bolt grips and lengths (continued).

Grip Number	-9 Diameter		-10 Diameter		-12 Diameter	
	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length
53	3.342	4.124	3.342	4.185	3.342	4.342
54	3.405	4.186	3.405	4.248	3.405	4.405
55	3.468	4.248	3.468	4.311	3.468	4.468
56	3.530	4.311	3.530	4.373	3.530	4.350
57	3.592	4.374	3.592	4.435	3.592	4.592
58	3.655	4.436	3.655	4.498	3.655	4.655
59	3.718	4.499	3.718	4.561	3.718	4.718
60	3.780	4.561	3.780	4.623	3.780	4.780
61	3.843	4.624	3.843	4.689	3.843	4.843
62	3.905	4.686	3.905	4.748	3.905	4.905
63	3.968	4.749	3.968	4.811	3.968	4.968
64	4.030	4.811	4.030	4.873	4.030	5.030
65	4.093	4.874	4.093	4.936	4.093	5.093
66	4.155	4.936	4.156	4.998	4.155	5.155
67	4.218	4.999	4.218	4.061	4.218	5.218
68	4.280	5.061	4.280	5.123	4.280	5.280
69	4.343	5.124	4.343	5.186	4.343	5.343
70	4.405	5.186	4.405	5.248	4.405	5.405
71	4.468	5.249	4.468	5.311	4.468	5.468
72	4.530	5.311	4.530	5.373	4.530	5.530
73			4.593	5.434	4.593	5.593
74			4.655	5.498	4.655	5.655
75			4.718	5.561	4.718	5.718
76			4.780	5.623	4.780	5.780
77			4.843	5.686	4.843	5.843
78			4.905	5.748	4.905	5.905
79					4.968	5.968
80					5.030	6.030
81					5.093	6.093
82					5.155	6.155
83					5.218	6.218
84					5.280	6.280
85					5.343	6.343
86					5.405	6.405
87					5.468	6.468
88					5.530	6.530
89					5.593	6.593
90					5.655	6.655

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b. The bolt assemblies identified herein are proprietary products and are covered by the following patents: U.S. 4,087,896 (expiring May 9, 1997), 4,106,036 (expiring July 25, 1997), 4,048,898 (expiring September 20, 1996), Great Britain 1,524,866, Canada 1,023,585, Japan 50-125256, Italy 1,051,593; Israel 515578; and Germany 2,546,641.5. The Government does not have a royalty-free license.

Custodians:
Army - AV
Navy - AS
Air Force - 99

Review:
Air Force - 82

Preparing Activity:
Navy - AS
Project No. 5306-1371

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